Manual

HYDRA Information Interface for SAP PP

SAP-ISS 8.1

Version 1.1.19800

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Contents

1 Information Interface for SAP PP

Use options

The information interface for SAP PP allows for the download of additional data (e.g. long text, component data), which are not included in the PP-PDC interface used to link MES to SAP.

This enables the provision of additional data and also of customer-specific tables in SAP to the MES, if necessary.

Implementation notes

Use the information interface for SAP PP if:

- If you use the PP-PDC-interface for the transfer of production orders/ production transaction data from SAP to the MES and if you also wish to transfer data that are not included there to the MES.

Integration

If you use the SAP-PPPDC component, the added orders/ operations form the reporting structure for numerous additional components in HYDRA.

The data transferred like that may be completed by the information interface SAP PP in order to provide data for the use of additional function packages.

Communication component to transfer provided additional information from SAP R/3 to the HYDRA data base

Integration into the SAP business framework by synchronous RFC (Remote Function Call)

Assurance of data consistency during the integration into the PP-PDC environment.

Flexible transfer of additional information in the provided format

Requirements: the additional information is provided according to the MPDV specifications. This function module is implemented under the customer's responsibility or is realized as an option in the context of a special service order in connection with MPDV.

Scope of functions

- Communication component to transfer provided additional information from SAP R/3 and/or ECC to the HYDRA database
- Integration into the SAP business framework by synchronous RFC (Remote Function Call)

- Assurance of data consistency in the integration into the PP-PDC environment.
- Flexible transfer of additional information in the provided format
- Requirements:
  - the additional information is provided according to the MPDV specifications.

On the SAP side, this function module is implemented under the customer's responsibility or is realized as an option in the context of a special service order in connection with MPDV.

2 Data type definitions

Type	Description
CHAR x	Information is left-aligned for the data type CHAR. Places that are not required are filled with blanks (U+0020). If the field is not used, it must be completely prepopulated with blanks. Example: "ABCD "
NUM x	Numeric field of the length x without sign. The data type NUMC only supports digits (ASCII characters 30 Hex to 39 Hex). These digits are right-aligned and unnecessary places are filled with zeros. If the field is not used, it must be completely prepopulated with zeros (U+0030). Example: "00000002"
DEC x.y **QUAN x.y**	Numeric field of the length x and y decimal places. A data field in HYDRA format is **preceded** by a sign ("+" or "-") and includes a decimal point. Places that are not required are filled with zeros. If the field is not used, it must be completely prepopulated with zeros (U+0030) including algebraic sign and decimal separator. e.g. DEC 13,3: • `-1234567890,123` → `-1234567890.123` • `234567890,3` → `+0234567890.300` Note: The field length is indicated WITHOUT algebraic sign and WITHOUT decimal point in the tabular description of the structure. This means, for example, that a field QUAN 13.3 is converted to an external length of CHAR15.
DATE	Format YYYYMMDD. If the field is not used, it must remain empty (filled with blanks (U+0020).
TIME	Format HHMMSS. If the field is not used, it must be set to “000000” (zeros with (U+0030)).

HYDRA does not support special characters for all alphanumeric fields. This includes, among others: "\“ (backslash), "|“ (pipe), „ “ “ (double quotes), and " ’ “ (single quotes). You cannot enter these characters using shop floor terminals and the MOC does not support them.

The character " **;** “ (semicolon) is used as a separator in the system. You must not use this character in key fields (e.g. order/operation number, MES batch number, personnel number).

The character " % " (percent) is used as a placeholder for database communication. You should not use this character to prevent the result from being falsified.

3 Interface Extension HYINFO

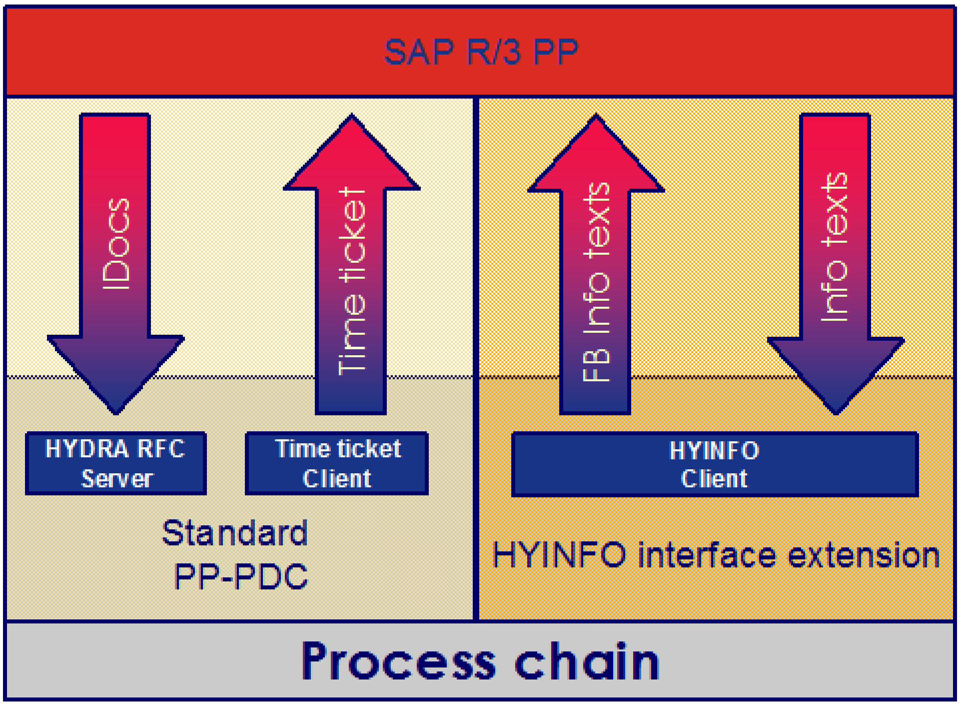
Overview

The fact that the PP-PDC and its predecessor (KK 2) were that often integrated into the process chain of numerous organizations shows that in addition to the information provided by default message types there is also a strong demand for further information on orders and operations. This is the reason why you can additionally implement the PP-PDC interface extension developed by MPDV to meet the customers' requirements.

Implement the RFC client "HYINFO“ in the MES to establish the PP-PDC extension. The MES processing specifies the required structures. Moreover a function module is integrated into SAP. This function module selects the data from the SAP data model. The HYINFO client calls this function module.

You can also extend the information interface to meet upcoming demands and to integrate additional business units. This means that the function module can easily be customized to specific requirements.

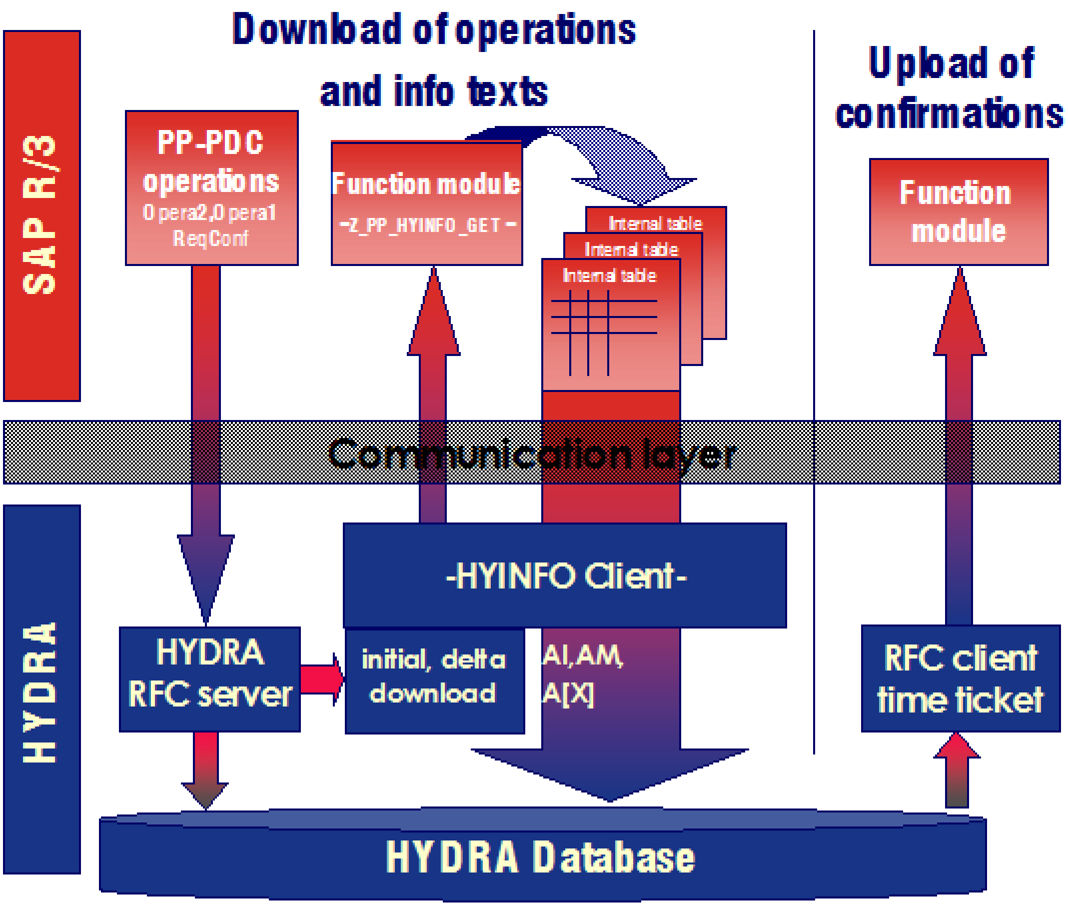
The following presentation illustrates the data paths in a possible customer solution.



The HYINFO client

Initial and delta downloads transfer operation data to HYDRA via the PP-PDC interface. In order to select the necessary additional data for these operations, the HYDRA RFC server starts the responsible RFC client. This RFC client transfers the operation data from the Opera2-IDoc to the function module in R/3. Then the function module transmits the additional data to HYINFO. This communication takes place synchronously.

The base structure of the internal table is based on the HYDRA structure "Additional information on the operation" (HYINFO). It is the same structure for all used information records. There is only one difference in the assignment of the fields related to their origin in SAP and to their usage in MES.



The function module "Z_PP_HYINFO_GET"

The function module to be implemented in SAP is developed in the customer's SAP namespace (in most cases the "Z" namespace).

The transfer of the record types from this module, can offer both, be geared to the HYINFO record types and include customer- and/or project-specific record types. How this data is transferred and used in the MES must be defined in each individual project.

You only have to transfer those segments that include the fields you need. For example, if you only require header data, you only have to transfer the record type "AK" but no HYINFO-AV segments.

MPDV may provide a sample implementation (as text file; cannot directly be implemented in SAP). If necessary, please contact your Project Manager.

Module definition:

Function module		***Z_PP_HYINFO_GET***
IMPORT		
	- none -	
EXPORT		
	- none -	
TABLES		
	Parameter name	**Type**
	ZPP_HYINFO	ZPP_HYINFO
	ZPP_HYBAPI[1]	ZPP_HYBAPI

Input data of the table ZPP_HYINFO

This internal table transfers those operations for which additional data is to be added.

The table has the same structure as for the return of the data. Only the following key data fields are completed:

- ORDERID
- SEQUENCE
- OPERATION
- SUB_OPER
- SUBSYSTEM_GROUPING

[1] Processing for table ZPP_HYBAPI is available from program version hysapinf.exe/out V8.1.1.43.

Input data of the table ZPP_HYBAPI

The table ZPP_HYBAPI does not require input data.

Output data of the table ZPP_HYINFO

The function module returns the data to be added in an internal table.

This table has the following structure (***ZPP_HYINFO):***

Field name	T	L	Meaning
ORDERID ☒	CHAR	12	Order
SEQUENCE ☒	CHAR	6	Sequence
OPERATION ☒	CHAR	4	Operation
SUB_OPER ☒	CHAR	4	Suboperation
SUBSYSTEM_GROUPING ☒	CHAR	3	Grouping, subsystem (from SAP default data)
RECORDTYPE ☒	CHAR	2	Information type ("AI", "AM", "AK", "AX", "AU")
PAGENO	NUMC	8	Serial numbering within a key (marked by ☒) ... starting with "00000001"
INFOTEXT1	CHAR	80	Additional information text 1
INFOTEXT2	CHAR	80	"
INFOTEXT3	CHAR	80	"
INFOTEXT4	CHAR	80	"
INFOTEXT5	CHAR	80	"
INFOTEXT6	CHAR	80	"
INFOTEXT7	CHAR	80	"
INFOTEXT8	CHAR	80	"
INFOTEXT9	CHAR	80	"
INFOTEXT10	CHAR	80	"

Output data of the table ZPP_HYBAPI

The function module returns the data to be added in an internal table. The internal table includes HYDRA dialog data strings. These dialog data strings are specified as part of the implementation project.[2]

Field	T	L	D	Description
Transaction	CHAR	20		Transaction ID (dialog ID in HYDRA) - no functional significance
Description	CHAR	40		Plain text description as comment - no functional significance
Data	CHAR	940		Dialog data string for HYDRA in the PDM format. Example: ANR.MODIFY|ANR.ANR=08150010|MNR=Maschine1|ATK=ABC| …

[2] Processing for table ZPP_HYBAPI is available from program version hysapinf.exe/out V8.1.1.43.

Column legend

Column	Description
Field name	Name of the field
V (usage)	S Key field clearly identifying the data record. (Further key fields might be required). The field must be completed. M Mandatory field which must be filled with a valid value. ML Mandatory field if the HYDRA Shop Floor Scheduling is used (HLS). MM Mandatory field if the HYDRA Material and Production Logistics (MPL and/or MPL/RF) is used. K Field may stay empty (optional field). SA Mandatory field if the Arburg control system (SCS-ALS) is in use; otherwise optional field.
T(type)	Data type of the field
L(ength)	Field length For fields of data type DEC/QUAN: Overall number of digits without decimal separator and algebraic sign.
D(ecimal places)	For fields of data type DEC/QUAN: Number of decimal places; otherwise: not relevant.
Description	Field description and/or comment on the field.

Order header data (AK)

Field name	V	T	L	D	Description	Usage in HYDRA
ORDERID	M	CHAR	12	0	Order	
SEQUENCE	M	CHAR	6	0	Sequence	
OPERATION	M	CHAR	4	0	Operation	
SUB_OPER	M	CHAR	4	0	Suboperation	
SUBSYSTEM_GROUPING	M	CHAR	3		Subsystem grouping	
RECORDTYPE	M	CHAR	2		HYINFO record type Fixed "AK"	
PAGENO	M	NUM	8		Fixed "00000001"	
INFOTEXT1	M	QUAN	13	3	Base quantity	Order information → Order header (AK) → General → Target quantity In general, the SAP order header quantity is integrated into the HYDRA basic quantity. Deviant definitions are project-specific and are explicitly presented as such.
	M	CHAR	3		Base quantity unit	Order information → Order header (AK) → General → Target quantity unit
	K	QUAN	13	3	Order scrap quantity	Order information → Order header (AK) → General → Estimated scrap
	K	QUAN	13	3	Order header quantity	Not used
INFOTEXT2	ML	DATE			Order end date	Order information → Order header (AK) → Dates → Order end date

Field name	V	T	L	D	Description	Usage in HYDRA
	ML	TIME			Order end time Please note: HYDRA does not support 240000; transfer 235959 instead.	Order information → Order header (AK) → Dates → Order end time
	ML	DATE			Order start date	Order information → Order header (AK) → Dates → Order start date
	ML	TIME			Order start time Please note: HYDRA does not support 240000; transfer 235959 instead.	Order information → Order header (AK) → Dates → Order start time
	ML	DATE			Scheduled end (date)	Order information → Order header (AK) → Dates → Scheduled end
	ML	TIME			Scheduled end (time)	Order information → Order header (AK) → Dates → Scheduled end
	ML	DATE			Scheduled start (date)	Order information → Order header (AK) → Dates → Scheduled start
	ML	TIME			Scheduled start (time)	Order information → Order header (AK) → Dates → Scheduled start
INFOTEXT3	K	CHAR	10		Sales order	Order information → Order header (AK) → General → Sales order
	K	CHAR	6		Sales order item	Order information → Order header (AK) → General → Sales order item
	K	CHAR	3		MRP controller	Order information → Order header (AK) → Assignment → MRP controller
	K	CHAR	3		Production controller	Order information → Order header (AK) → Assignment → Order group
	K	CHAR	8		Task list group	Order information → Order header (AK) → Assignment → Work plan
	K	CHAR	2		Group counter	Order information → Order header (AK) → Assignment → Work plan version
INFOTEXT4	M	CHAR	18		Material number	Order information → Article
	M	CHAR	40		Material name	Order information → Order header (AK) → General → Article description
	M	CHAR	5		Order type	Order information → Order type Fixed "0“, if not project-specific.
	K	CHAR	15		eKANBAN control cycle	eKANBAN control cycle This field is available as of the MLE version HY72_030.

Field name	V	T	L	D	Description	Usage in HYDRA
INFOTEXT5	K	CHAR	40		Customer name/designation	Order information → Order header (AK) → Customer name This field is available as of the MLE version HY72_032.
INFOTEXT6	K	CHAR	50		Drawing issue number	Order information → Order header (AK) → Drawing issue number This field is available as of BDE82 and MLE version HY72_032.

User fields for the header/ operation (AU)

Both, at the order header and at the operation, HYDRA provides 66 user fields, respectively, of different data types. **Only** use the record type "AU" to transfer the user fields for the order header and the operation. The data record itself differentiates between order header and operation.

Assign the key fields as follows to ensure correct data is transferred to HYDRA:

Field name	User fields for the order header	User fields for the operation
ORDERID	SAP order number	SAP order number
SEQUENCE	Empty	SAP sequence number
OPERATION	Empty	SAP operation number
SUB_OPER	Empty	SAP suboperation number
RECORDTYPE	“AU“	“AU“
INFOTEXT10.Satzart	“AU“	“AG“

The complete data record structure:

Field name	V	T	L	D	Description	Usage in HYDRA

Field name	V	T	L	D	Description	Usage in HYDRA
ORDERID	M	CHAR	12	0	Order	SAP order number - for order header - for operation)*
SEQUENCE	M	CHAR	6	0	Sequence	SAP order number - for operation)*
OPERATION	M	CHAR	4	0	Operation	SAP order number - for operation)*
SUB_OPER	M	CHAR	4	0	Suboperation	SAP order number - for operation)*
SUBSYSTEM_GROUPING	M	CHAR	3		Subsystem grouping	
RECORDTYPE	M	CHAR	2		HYINFO record type (AU)	
PAGENO	M	NUM	8		Fixed "00000001"	
INFOTEXT1	K	CHAR	8		User field key	USRFLD Fixed "SYSTEM“, if not project-specific
	K	DATE	8		User field 1	FU:1
	K	DATE	8		User field 2	FU:2
	K	DATE	8		User field 3	FU:3
	K	DATE	8		User field 4	FU:4
	K	DATE	8		User field 5	FU:5
	K	DATE	8		User field 6	FU:6
	K	NUM	8		User field 7	FU:7
	K	CHAR	16		Filler	Not used
INFOTEXT2	K	NUM	8		User field 8	FU:8
	K	NUM	8		User field 9	FU:9
	K	NUM	8		User field 10	FU:10
	K	NUM	8		User field 11	FU:11
	K	NUM	8		User field 12	FU:12
	K	NUM	8		User field 13	FU:13
	K	NUM	8		User field 14	FU:14
	K	NUM	8		User field 15	FU:15
	K	NUM	8		User field 16	FU:16
	K	NUM	8		User field 17	FU:17
INFOTEXT3	K	NUM	8		User field 18	FU:18
	K	NUM	8		User field 19	FU:19
	K	NUM	8		User field 20	FU:20
	K	NUM	8		User field 21	FU:21
	K	NUM	8		User field 22	FU:22
	K	DEC	13	3	User field 23	FU:23
	K	DEC	13	3	User field 24	FU:24
	K	CHAR	10		Filler	Not used
INFOTEXT4	K	DEC	13	3	User field 25	FU:25
	K	DEC	13	3	User field 26	FU:26
	K	DEC	13	3	User field 27	FU:27
	K	DEC	13	3	User field 28	FU:28
	K	CHAR	1		User field 29	FU:29
	K	CHAR	1		User field 30	FU:30
	K	CHAR	1		User field 31	FU:31
	K	CHAR	1		User field 32	FU:32

Field name	V	T	L	D	Description	Usage in HYDRA
	K	CHAR	1		User field 33	FU:33
	K	CHAR	1		User field 34	FU:34
	K	CHAR	1		User field 35	FU:35
	K	CHAR	1		User field 36	FU:36
	K	CHAR	1		User field 37	FU:37
	K	CHAR	1		User field 38	FU:38
	K	CHAR	1		User field 39	FU:39
	K	CHAR	1		User field 40	FU:40
	K	CHAR	1		User field 41	FU:41
	K	CHAR	1		User field 42	FU:42
	K	CHAR	1		User field 43	FU:43
	K	CHAR	1		User field 44	FU:44
	K	CHAR	4		Filler	Not used
INFOTEXT5	K	CHAR	10		User field 45	FU:45
	K	CHAR	10		User field 46	FU:46
	K	CHAR	10		User field 47	FU:47
	K	CHAR	10		User field 48	FU:48
	K	CHAR	10		User field 49	FU:49
	K	CHAR	10		User field 50	FU:50
	K	CHAR	20		User field 51	FU:51
INFOTEXT6	K	CHAR	20		User field 52	FU:52
	K	CHAR	20		User field 53	FU:53
	K	CHAR	20		User field 54	FU:54
	K	CHAR	20		User field 55	FU:55
INFOTEXT7	K	CHAR	20		User field 56	FU:56
	K	CHAR	20		User field 57	FU:57
	K	CHAR	20		User field 58	FU:58
	K	CHAR	20		User field 59	FU:59
INFOTEXT8	K	CHAR	20		User field 60	FU:60
	K	CHAR	20		User field 61	FU:61
	K	CHAR	20		User field 62	FU:62
	K	CHAR	20		User field 63	FU:63
INFOTEXT9	K	CHAR	20		User field 64	FU:64
	K	CHAR	40		User field 65	FU:65
	K	CHAR	20		Filler	Not used
INFOTEXT10	K	CHAR	40		User field 66	FU:66
	M	CHAR	2		Record type	"AU" Order header "AG" OP / Operation

Operation data (AV)

Use the record type "AV" to transfer operation-related additional data from SAP to HYDRA. This includes several process times and controlling information referring to operations.

In doing so, the system already transfers individual fields via the PP-PDC structure OPERA2. Consequently, you can transfer these fields customer-specifically and thus deviate from the default processing method in both, SAP and HYDRA.

Field name	V	T	L	D	Description	Usage in HYDRA
ORDERID	M	CHAR	12	0	Order	
SEQUENCE	M	CHAR	6	0	Sequence	
OPERATION	M	CHAR	4	0	Operation	
SUB_OPER	M	CHAR	4	0	Suboperation	Not used
SUBSYSTEM_GROUPING	M	CHAR	3		Subsystem grouping	
RECORDTYPE	M	CHAR	2		HYINFO record type Fixed "AV“	
PAGENO	M	NUM	8		Number of pages Fixed "00000001"	
INFOTEXT1	ML	CHAR	20		HYDRA material type	You can use the HYDRA material type to control processing, in particular when the HYDRA MPL module is used. If the HYDRA MPL module is not used, the field must be transferred with the value "SYSTEM".
	K	CHAR	1		External priority	Priority (0 - 9; 9 = priority high)
	K	CHAR	20		Color	Color of the material
	K	CHAR	8		Cost center	Cost center of the operation
	K	CHAR	10		Cost type	
	M	CHAR	1		Authorization level of the operation	Authorization level to log in and off the OP (lowest authorization = 1)
	K	QUAN	13	3	Minimum send ahead quantity in the (primary) quantity unit of the operation	
	M	CHAR	5		Filler	Complete with blanks (fixed)
INFOTEXT2	M	NUMC	8		Processing time	Processing time in seconds
	K	NUMC	8		Teardown (retooling) time in seconds	
	K	NUMC	8		Delivery time in seconds	Delivery time in seconds At present only relevant in connection with the HYDRA Shop Floor Scheduling module (HLS). Should be set to 0 if this is not an external processing OP.
	ML	CHAR	1		Indicator external processing OP	At present only relevant in connection with the HYDRA Shop Floor Scheduling module (HLS). Should in general be set to "N".
	M	CHAR	6		Remaining run time formula	Only relevant, if you use the HYDRA Shop Floor Scheduling module (HLS). Transfer fixed "RLFZ" in this case. Deviating settings are possible if you customize HYDRA.

Field name	V	T	L	D	Description	Usage in HYDRA
	K	NUMC	8		Lead time	
	K	NUMC	8		Max. synchronization time in seconds	
	K	NUMC	8		Waiting time in seconds	
	K	NUMC	8		Minimum waiting time in seconds	
	K	NUMC	8		Wait time in seconds	
	K	CHAR	4		Wage type	
	K	CHAR	1		Piecework indicator/ premium	
	M	CHAR	4		Filler	Complete with blanks (fixed)
INFOTEXT3	K	QUAN	13	3	Premium default: te in seconds per 1000 pieces	
	K	QUAN	13	3	Premium default tr in seconds	
	K	QUAN	13	3	Premium default: teb in seconds per 1000 pieces	
	K	QUAN	13	3	Premium default trb in seconds	
	M	CHAR	6		Processing code	fixed "SYSTEM" Deviating settings are possible if you customize HYDRA.
	M	NUMC	8		Target cycle	Target cycle in seconds/ 1000; should be set; mandatory for MDE cycle monitoring
	M	CHAR	6		Filler	Complete with blanks (fixed)
INFOTEXT4	M	NUMC	8		Partitioning	Partitioning or fixed "00000001"
	K	DEC	5	2	Machine/operator ratio, setup	
	K	DEC	5	2	Machine/operator ratio, production	
	M	QUAN	13	3	Target quantity of the operation	
	K	CHAR	10		Cost center	Alternative: Cost center of the operation with 10 digits
INFOTEXT5	M	DEC	13	3	Underdelivery in percent	Underdelivery in percent The value entered is an absolute percentage value of the target quantity (primary quantity unit). Example: - Target quantity of the operation: 120 pieces - Underdelivery: 84% The actual quantity must not fall below 101 items. This field is available as of the MLE version HY72_032.

Field name	V	T	L	D	Description	Usage in HYDRA	
	M	CHAR	1		Reaction to underdelivery Possible values: " " no reaction "W" Warning; entry of a deviation reason required "X" Error; underdelivery not allowed.	Reaction to underdelivery Possible values: " " no reaction "W" Warning; entry of a deviation reason required "X" Error; underdelivery not allowed. Note: You can only enter deviation reasons for overdeliveries/ underdeliveries via the CTWIN software. If you use DOS terminals, the reaction "W" is interpreted as an error ("X"). This field is available as of the MLE version HY72_032.	
	M	DEC	13	3	Overdelivery in percent	Overdelivery in percent The value entered is an absolute percentage value of the target quantity (primary quantity unit). Example: - Target quantity of the operation: 120 pieces - Overdelivery: 168% The actual quantity must not exceed 201 items. This field is available as of the MLE version HY72_032.	
	M	CHAR	1		Reaction to overdelivery Possible values: " " no reaction "W" Warning; entry of a deviation reason required "X" Error; overdelivery not allowed.	Reaction to overdelivery Possible values: " " no reaction "W" Warning; entry of a deviation reason required "X" Error; overdelivery not allowed. Note: You can only enter deviation reasons for overdeliveries/ underdeliveries via the CTWIN software. If you use DOS terminals, the reaction "W" is interpreted as an error ("X"). This field is available as of the MLE version HY72_032.	
	K	CHAR	1		Person OK	Staff OK This field is available as of the MLE version HY72_032.	ANR. TA:1
	K	CHAR	1		Tool OK	Tool OK This field is available as of the MLE version HY72_032.	ANR. TA:2
	K	CHAR	1		Material OK	Material OK This field is available as of the MLE version HY72_032.	ANR. TA:3
	K	NUM	8		PEP: Qualification: manufacturing	QUAL:NORM This field is available as of the MLE version HY72_032.	
	K	NUM	8		PEP: Qualification: setup	QUAL:RUE This field is available as of the MLE version HY72_032.	

Field name	V	T	L	D	Description	Usage in HYDRA
INFOTEXT6	K	CHAR	50		Article index	This field is available as of the BDE82 MLE version HY72_032.

Remarks on selected fields

If you use the record type "AV", the following fields of the structure "AV" must contain values: "processing time", "partitioning" or "target quantity of the operation".

Target cycle

1) The system tries to read the cycle time from the field "target cycle" (SOLLZYKLUS). If this is possible, the system enters this value as the target cycle time in the HYDRA operation.

2) If the system cannot identify the target cycle time according to 1), then the system calculates it from the specified processing time (BEARBEITUNGSZEIT) and the partitioning (TEILIGKEIT) values that might be specified.

Calculation basis (formulas):
Processing time OP = Target quantity OP * (target cycle time machine/ partitioning)
- Target cycle time machine (= BDE cycle time/ machine stroke)
- Target cycle time/ unit = target cycle time machine/ partitioning
=> Target cycle time machine= (processing time * partitioning)/ target quantity OP

Target partitioning

1) If the field "TEILIGKEIT" (partitioning) includes a value, the system enters this value as the partitioning in the operation.

2) If this is not the case, the system implicitly assumes a partitioning of 1 and enters this value in the operation.

Processing time

3) In HYDRA the content of the field "processing time" (PROCESS_TIME) specifies the standard time for machine assignment (target for RPA 11).

Underdelivery/overdelivery in %

1) The system transfers the values for underdeliveries/overdeliveries and the corresponding reactions from the E2BP_PP_PDC_OPERA2000 segment.

2) You can overwrite these values with customer-specific values in the HYINFO "AV". Enable an INI configuration if you want to prevent the values from being overwritten.

Additional texts for the operation (AI)

The structure "additional texts for the operation (AI)" offers numerous options to display additional information on an operation via the terminal. This includes dimensions, quality specifications or handling instructions. Once you have selected the operation, you can directly view the additional information in the terminal since the link between operation and → information was already established during the download from R/3.

Field name	V	T	L	D	Meaning	Usage in HYDRA
ORDERID	M	CHAR	12		Order	
SEQUENCE	M	CHAR	6		Sequence	
OPERATION	M	CHAR	4		Operation	
SUB_OPER☒	M	CHAR	4		Suboperation	
SUBSYSTEM_GROUPING	M	CHAR	3		Grouping, subsystem (from SAP default data)	
RECORDTYPE	M	CHAR	2		Type of information Fixed "AI"	
PAGENO	M	NUMC	8		Serial numbering within a key (marked by ☒) ... starting with "00000001"	
INFOTEXT1	K	CHAR	80		Additional information text 1	
INFOTEXT2	K	CHAR	80		"	
INFOTEXT3	K	CHAR	80		"	
INFOTEXT4	K	CHAR	80		"	
INFOTEXT5	K	CHAR	80		"	
INFOTEXT6	K	CHAR	80		"	
INFOTEXT7	K	CHAR	80		"	
INFOTEXT8	K	CHAR	80		"	
INFOTEXT9	K	CHAR	80		"	
INFOTEXT10	K	CHAR	80		"	

Component list (AM)

The record type "BOM" and/ or "component list" was developed to gather BOM information on the operation, i.e. the component lists from R/3 and to make it available to the HYDRA user. Possible scenarios are for example: HYDRA displays the information or other systems, such as warehouse management systems, process the data. If you also use the HYDRA MPL module, retrograde withdrawal (backflushing) and posting in R/3 are possible.

Field name	V	T	L	D	Description	Usage in HYDRA

Field name	V	T	L	D	Description	Usage in HYDRA		
ORDERID	M	CHAR	12	0	Order			
SEQUENCE	M	CHAR	6	0	Sequence			
OPERATION	M	CHAR	4	0	Operation			
SUB_OPER	M	CHAR	4	0	Suboperation			
SUBSYSTEM_GROUPING	M	CHAR	3		Subsystem grouping			
RECORDTYPE	M	CHAR	2		HYINFO record type (Fixed "AM")			
PAGENO	M	NUM	8		Number of pages			
INFOTEXT1	ML	CHAR	18		Material number	Order information → OP → Components → Material	40	57
INFOTEXT2	K	CHAR	30		Additional text 1	Order information → OP → Components → Description 1	120	149
	K	CHAR	30		Additional text 2	Order information → OP → Components→ Description 2	150	179
INFOTEXT3	ML	QUAN	13	3	Input quantity of material required to produce 1 part.	Order information → OP → Components → Input quantity	200	214
	ML	CHAR	3		Quantity unit	Order information → OP → Components → QU Input quantity	215	217
	K	QUAN	13	3	Requirement quantity	Order information → OP → Components → Required quantity (ERP)	218	232
	ML	CHAR	10		Number of the BOM item	Order information → OP → Components → Item	233	242
	K	CHAR	1		Indicator for fixed quantity	Internal administration	243	243
	K	CHAR	1		Indicator for retrograde withdrawal/backflush	Internal administration	244	244
	ML	CHAR	4		Plant	Internal administration	245	248
	ML	CHAR	3		Unit of requirements quantity	Unit of requirements quantity This field is available as of the MLE version HY72_032.	249	251
	ML	CHAR	10		HYDRA material type	HYDRA material type This field is available as of the MLE version HY72_032.	252	261
	ML	CHAR	1		HYDRA consumption indicator	HYDRA consumption indicator If not specified otherwise, assign "L" to this field. For components of the material type "I" assign "N" to the field. This field is available as of the MLE version HY72_032. "L" Consumption referring to batches (by default) "D" Discrete consumption "A" Anonymous consumption "U" Automatic consumption	262	262

Field name	V	T	L	D	Description	Usage in HYDRA		
INFOTEXT4	ML	CHAR	40		Material short text	Order information → OP → Components → Description	280	319

Production resources and tools (AF)

Use the record type "AF" to transfer production resources and tools to HYDRA. In this case it is not important whether they are stored in SAP as production resources and tools. HYDRA shows the data in the order information dialog in OP → PRT.

Please also note that the first production resource and tool transferred for an operation will also be shown in the order information in OP → General → Tool.

Field name	V	T	L	D	Description	Usage in HYDRA		
ORDERID	M	CHAR	12	0	Order			
SEQUENCE	M	CHAR	6	0	Sequence			
OPERATION	M	CHAR	4	0	Operation			
SUB_OPER	M	CHAR	4	0	Suboperation			
SUBSYSTEM_GROUPING	M	CHAR	3		Subsystem grouping			
RECORDTYPE	M	CHAR	2		HYINFO record type Fixed "AF“			
PAGENO	M	NUM	8		Number of pages			
INFOTEXT1	M	CHAR	18		Material number of the resource	Order information → OP → Components → PRT	40	57
	K	CHAR	40		Resource name	Order information → OP → Components → PRT	58	97
INFOTEXT2	K	CHAR	30		Comment 1	Order information → OP → Components → PRT	120	149
	K	CHAR	30		Comment 2	Order information → OP → Components → PRT	150	179
	M	QUAN	13	3	Input quantity	Order information → OP → Components → PRT	180	194
	M	CHAR	3		Unit of the input quantity	Order information → OP → Components → PRT		
	M	CHAR	2		Filler	Not used		
INFOTEXT3	M	CHAR	4		Resource type Fixed "WNR"	Order information → OP → Components → PRT	200	203

Documents (AC)

Use the record type "AC" to transfer document links to HYDRA. If you can access the file system, the terminal displays these documents. In addition, the transferred document links will be displayed in the order information → OP → PRT.

Field name	V	T	L	D	Description	Usage in HYDRA
ORDERID	M	CHAR	12	0	Order	
SEQUENCE	M	CHAR	6	0	Sequence	
OPERATION	M	CHAR	4	0	Operation	
SUB_OPER	M	CHAR	4	0	Suboperation	Not used
SUBSYSTEM_GROUPING	M	CHAR	3		Subsystem grouping	
RECORDTYPE	M	CHAR	2		HYINFO record type Fixed "AC“	
PAGENO	M	NUM	8		Number of pages	
INFOTEXT1	M	CHAR	18		Material number/ identification of the document/ graphic	Order information → OP → Components → PRT **Please note:** The material number/ identification must be unique in an operation; if several document links are transferred they must include different values.
	K	CHAR	40		Name	Order information → OP → Components → PRT
INFOTEXT2	K	CHAR	30		Additional name 1 (option)	Order information → OP → Components → PRT
	K	CHAR	30		Additional name 2 (option)	Order information → OP → Components → PRT
	M	QUAN	13	3	Input quantity Fixed "1.00"	Order information → OP → Components → PRT
	M	CHAR	3		Unit of the input quantity Fixed "ST" (pieces/pcs.)	Order information → OP → Components → PRT
	M	CHAR	2		Filler	Not used
INFOTEXT3	M	CHAR	8		Path	Order information → OP → Components → PRT Reference to a path that is defined in the path configuration (Menu *File > System administration > Paths*).
	M	CHAR	72		File name incl. file extension	Order information → OP → Components → PRT
INFOTEXT4	M	CHAR	56		Continuation of the file name incl. file extension (if field in INFOTEXT3 is not sufficient)	Order information → OP → Components → PRT
	M	CHAR	4		Resource type Fixed "DOC"	Order information → OP → Components → PRT
	M	CHAR	20		Filler	Not used
INFOTEXT5	K	CHAR	25		Alternative: Identification of the document/ graphic (document ID) with 25 digits	Order information → OP → Components → PRT **Please note:** The material number/ identification must be unique in an operation; if several document links are transferred they must include different values.

Unlock operation for processing (AE)

Use the record type "AE" to unlock the operation after interface processing, if you locked the operation beforehand using the INI configuration.

If the system transfers the segment with the record type "AE" and the INI configuration that locks operations is enabled for the HYINFO processing, then the transferred operation will be unlocked and available for posting in the system.

Go to the Order information → OP → Administration → "Locked interface“ to view if an operation is locked for HYINFO processing.

Field name	V	T	L	D	Description	Usage in HYDRA
ORDERID	M	CHAR	12	0	Order	
SEQUENCE	M	CHAR	6	0	Sequence	
OPERATION	M	CHAR	4	0	Operation	
SUB_OPER	M	CHAR	4	0	Suboperation	Not used

When to use this function?

Enable the automatic lock using the INI configuration, if you want to make sure that an operation can only be further processed in the system, once all required data and additional data have been transferred via the interface.

How does this function work?

If the INI configuration is available and enabled, the system locks the relevant operation when transferring the additional data via the HYINFO module in the OPERA2000 segment. HYDRA automatically locks the operation. Transfer the "AE" segment if you want to unlock the operation after HYINFO processing. Transfer the "AE" segment explicitly for each operation.

You cannot log on the operation to the AIP terminal as long as the operation is locked by the interface. If you attempt to log on the operation to the terminal, the system rejects the logon with the error code 73 "OP is locked by MLE interface".

This segment is available as of MLE variant HY72_034. For further information please refer to documentation HYDRA Settings Relevant to the Application

4 HYDRA Settings Relevant to the Application

	→	

Maintenance of the HYDRA distribution model – inbound

Change existing entries or create new entries in the HYDRA distribution model for HYDRA inbound processing:

Parameter name	Value
Processing the PPCC2RECORDER IDoc of the PP-PDC interface	
Message type	PPCC2RECORDER
Priority	None
Command	hysapinf.scr
Command parameter	/RECTYPE
Description	PP-PDC / HYINFO – request additional data
Log. target system	Created logical system
Retention period	10
Processing the PPCC2RECORDER IDoc and HYINFO additional data	
Message type	PPCC2HYINFOORDER
Priority	None
Command	mle72imp.scr
Command parameter	/VARIANTE=<MLE version to be used>
Description	PP-PDC / HYINFO transfer of additional data
Log. target system	Created logical system
Retention period	10

No deletion of the operation's material, tool, NC program

If HYDRA receives components, tools, or NC programs for the operation, the system enters the first element transferred for the operation in the corresponding fields "material", "tool" or "DNC". The same applies if the tool is assigned by a production variant/version in the Shop Floor Scheduling module.

In both cases it might be necessary to prevent these operation entries from being deleted when the components and PRT lists are deleted routinely while the system transfers the next operation. You can achieve this by setting the relevant entries in the HYDRA INI configuration:

Parameter name	Value
For the material	
INI name	SAP
Section	E2BP_PP_PDC_OPERA2000
Key	PREVENT_ANR_RES_MAT_FROM_DELETION
Value	Y → yes N → no
Active	Yes
Comment	Prevents the first resource of the type "material" (ID "MAT") included in the component list from being deleted when the component list is deleted.
For the tool	
INI name	SAP
Section	E2BP_PP_PDC_OPERA2000
Key	PREVENT_ANR_RES_WNR_FROM_DELETION
Value	Y → yes N → no
Active	Yes

Parameter name	Value
Comment	Prevents the first resource of the type "tool" (ID "FHM"/"PRT") included in the component list from being deleted when the PRT list is deleted.
For the DNC program	
INI name	SAP
Section	E2BP_PP_PDC_OPERA2000
Key	PREVENT_ANR_RES_DNC_FROM_DELETION
Value	Y → yes N → no
Active	Yes
Comment	Prevents the first resource of the type "DNC resource" (ID "DNC") included in the component list from being deleted when the component list is deleted.

Requirements:

.\custom\userexit\mle_convfield_in.hsc:	V8.1.1.64407
MLE version:	HY72_031

Preventing the call for deletion records

Use an INI entry if you do not want to request additional data for deletion records:

Parameter name	Value
INI name	SAP
Section	REQUEST_DELETED_OP
Key	PPCC2RECORDER for PP-PDC

Parameter name	Value
	OPERA3 for KK3 OPERA4 for KK4
Value	Y → yes N → no
Active	Yes
Comment	No additional data for deletion records

Requirements:

hysapinf.exe/out: V8.1.1.41

Preventing the transfer of overdelivery/underdelivery details from record type "AV"

Configure the HYDRA INI configuration to prevent the overdelivery/underdelivery information from the record type "AV" from being transferred:

Parameter name	Value
INI name	SAP
Section	HYINFO_AV
Key	USE_OVER_UNDER_DELIVERY_CHECK
Value	Y → yes (set by default) N → no
Active	Yes
Comment	Transfers overdelivery/underdelivery data from the HYINFO AV segment

If you use the information interface for SAP PP, you cannot use BAPINOUPDATE for the fields of the following record types:

- Order header data (AK)
- User fields for the header/ operation (AU)
- Operation data (AV)
- Additional texts for the operation (AI)
- Component list (AM)
- Production resources and tools (AF)
- Documents (AC)

Activating the transfer of dialog data strings

Use the following program parameter to activate the transfer of dialog data strings via the HYINFO module in the ZPP_HYBAPI structure when calling the executing program in the script hysapinf.scr:

Call prior to the change (example for Windows):

```
$HY_PATH/hysapinf.exe /TL=TRL_ALL $1 $2 $3 $4 $5 $6 $7
```

Call after the change (example for Windows):

```
$HY_PATH/hysapinf.exe /Z2BAPI000=ADD $1 $2 $3 $4 $5 $6 $7
```

Requirements:

hysapinf.exe/out:	V8.1.1.43

Changing the name of the function module Z_PP_HYINFO_GET

If you cannot use the name of the HYINFO function module given by MPDV (e.g. as internal naming conventions prevent it), you can change this name. To do so, use the following program parameter when calling the executing program in the script hysapinf.scr:

Call prior to the change (example for Windows):

```
$HY_PATH/hysapinf.exe /TL=TRL_ALL $1 $2 $3 $4 $5 $6 $7
```

Call after the change (example for Windows):

```
$HY_PATH/hysapinf.exe /HYINFO_GET=<New Name> $1 $2 $3 $4 $5 $6 $7
```

Changing the name of the table ZPP_HYINFO

If you cannot use the name of the ZPP_HYINFO table given by MPDV (e.g. as internal naming conventions prevent it), you can change this name. To do so, use the following program parameter when calling the executing program in the script hysapinf.scr:

Call prior to the change (example for Windows):

```
$HY_PATH/hysapinf.exe /TL=TRL_ALL $1 $2 $3 $4 $5 $6 $7
```

Call after the change (example for Windows):

```
$HY_PATH/hysapinf.exe /HYINFO_ITAB=<New Name> $1 $2 $3 $4 $5 $6 $7
```

Changing the name of the table ZPP_HYBAPI

If you cannot use the name of the ZPP_HYBAPI table given by MPDV (e.g. as internal naming conventions prevent it), you can change this name. To do so, use the following program parameter when calling the executing program in the script hysapinf.scr:

Call prior to the change (example for Windows):

```
$HY_PATH/hysapinf.exe /TL=TRL_ALL $1 $2 $3 $4 $5 $6 $7
```

Call after the change (example for Windows):

```
$HY_PATH/hysapinf.exe /Z2BAPI_ITAB=<New Name> $1 $2 $3 $4 $5 $6 $7
```

Prevent processing of OPs until HYINFO is processed

Add the following entry to the HYDRA INI configuration if you want to lock operations until the original data and additional data have been imported completely from the HYINFO module (MBL_SAP_Implementation_HYINFO_FB).

Parameter name	**Value**
INI name	SAP
Section	E2BP_PP_PDC_OPERA2000
Key	SET_OPS_TO_BLOCKED
Value	Y → yes
Active	Yes

Parameter name	Value
Comment	Locks OPs until HYINFO has been processed

Requirements:

This function requires service pack 14 (or higher) or installation of an update package. The update package must include the following software components:

hymw.exe/out:	V8.1.1.646
hyerror.msg/en	V8.1.1.646
lib\b_anr.dll/so	V8.1.1.356
MLE version	HY72_034
MOC Language Resources	1.0.STD.66720